
Problem 1

This problem uses the Composite Trapezoidal rule and Composite Simpson's rule to approximate a definite integral of functions.

```
n = 6;
x = linspace(exp(1), exp(1)+2, n+1);
f = @(x) 1./(x.*log(x));

T = CompositeTrapezoidalRule(x, f(x));
S = CompositeSimpsonsRule(x, f(x));

% Exact value
I_exact = log(log(exp(1)+2));

fprintf('Composite Trapezoidal Rule: %.6f\n', T)
fprintf('Composite Simpson's Rule: %10.6f\n', S)
fprintf('Exact Value: %10.6f\n\n', I_exact)

% Errors
fprintf('Absolute Error (Trapezoidal): %.6e\n', abs(T - I_exact))
fprintf('Absolute Error (Simpson): %.6e\n', abs(S - I_exact))

%=====Functions=====
function approx = CompositeTrapezoidalRule(x, f)
%uses the Composite Trapezoidal rule to approximate the value of the
%definite integral given vectors of evenly spaced x values and
%corresponding function values

%INPUT
% x: vector of evenly spaced x values
% f: vector of corresponding function values

%OUTPUT
% approx: approximation for the definite integral

%defines step size
h = (x(end)-x(1))/(length(x)-1);

%calculates approximation using the Composite Trapezoidal rule
approx = h/2*(f(1)+2*sum(f(2:end-1))+f(end));
end

%
function approx = CompositeSimpsonsRule(x, f)
%uses the Composite Simpson's rule to approximate the value of the definite
%integral given vectors of evenly spaced x values and corresponding
%function values

%INPUT
% x: vector of evenly spaced x values
% f: vector of corresponding function values
```

```
%OUTPUT
% approx: approximation for the definite integral

%defines step size
h = (x(end)-x(1))/(length(x)-1);

%calculates approximation using the Composite Simpson's rule
approx = 1/3*h*(f(1)+4*sum(f(2:2:end-1))+2*sum(f(3:2:end-1))+f(end));
end
%=====

Composite Trapezoidal Rule: 0.441242
Composite Simpson's Rule: 0.439225
Exact Value: 0.439187

Absolute Error (Trapezoidal): 2.054971e-03
Absolute Error (Simpson): 3.867811e-05

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